

How Much of Reported Skill Is the Reference Forecast? A Controlled Evaluation for Short-Term Photovoltaic Power Forecasting

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Abstract—Photovoltaic power forecasting studies report large accuracy gains under widely varying protocol: of 27 recent papers we coded, 26 report no skill score against a reference forecast and all report point estimates without run-to-run variance. Using a leakage-controlled protocol on three co-located arrays (900 recorded runs, five architectures, five seeds), we show the reference forecast determines both the magnitude and the shape of reported skill. Against smart persistence, skill rises monotonically with horizon, read as a model’s advantage growing at longer lead times; against the optimal convex combination of climatology and persistence, the same forecasts give non-monotonic skill peaking at three hours. Night hours deflate normalised error by approximately 34 percent, closed-form in the daylight fraction, while skill is almost unchanged. Reported accuracy here is shaped more by the reference forecast and error metric than by the model: the largest difference across five architectures is 0.024 in skill score. All directions hold on a held-out test year.

Index Terms—Photovoltaic power forecasting, forecast verification, skill score, reproducibility, benchmark evaluation.

I. INTRODUCTION

Short-term photovoltaic (PV) power forecasting is an active research area, and reported accuracy improvements for hybrid deep-learning architectures are substantial. We built an open, leakage-controlled benchmark harness for this task on a public dataset: five architectures, evaluated across three module technologies, three forecast horizons, two feature regimes and five random seeds, with 900 machine-readable run records that reproduce every table and figure in this paper.

To characterise how such gains are currently reported, we coded 27 papers on hybrid PV power forecasting, published between 2022 and 2026, on eight dimensions of evaluation protocol, each judgement supported by a verbatim quotation from the source. Two counts summarise the picture: 26 of 27 report no skill score against any reference forecast, and all 27 report point estimates with no measure of run-to-run variance. Reported numbers in this literature are therefore difficult to compare with one another, or with a naive forecast, without knowing what they were measured against.

This paper measures what that comparison is worth. Holding the model, data and training procedure fixed, we vary only the reference forecast and the treatment of night hours. On one array at a six-hour horizon, the same forecasts score 0.652 against smart persistence and 0.194 against the optimal

convex combination of climatology and persistence recommended by Yang et al. [1] - 70 percent of the apparent skill is attributable to the reference. More consequentially, skill against persistence rises monotonically with horizon, the pattern usually read as a model’s advantage growing at longer lead times, while skill against the proper reference is non-monotonic and peaks at three hours: the trend is a property of the reference, not the model. Including night hours deflates normalised RMSE by approximately 34 percent, a factor that is closed-form in the daylight fraction, while the skill score is nearly unchanged. Across five architectures under an identical protocol, the largest difference we measure is 0.024 in skill score - smaller than either protocol effect.

A. Contributions

1. An open, leakage-controlled benchmark harness for short-term PV power forecasting, with 900 machine-readable run records reproducing every reported number.
2. A measurement of two evaluation-protocol effects - reference forecast and night-hour inclusion - held against a fixed model and data, and shown to exceed the largest architectural difference we measure.
3. A coded survey of 27 papers characterising current reporting practice on reference forecasts and run-to-run variance.

II. RELATED WORK

Hybrid architectures combining convolutional and recurrent layers, often with a further decomposition or gradient-boosted stage, dominate recent work on short-term PV power forecasting, and reported accuracy gains are large: a meta-analysis of 180 studies published since 2007 concludes that hybrid models consistently outperform alternatives [2]. That conclusion aggregates reported errors; this paper asks what those errors are measured against.

To characterise current practice, we coded 27 papers on hybrid PV power forecasting published between 2022 and 2026, on eight dimensions of evaluation protocol, each judgement supported by a verbatim quotation from the source (Table I). Beyond the two counts given in Section I, seventeen of 27 papers compare only against their own architecture’s components, and seventeen do not state whether night hours are excluded from evaluation - the two protocol axes this paper measures.

TABLE I

LITERATURE SURVEY OF EVALUATION PRACTICE, $n = 27$ PAPERS ON HYBRID CONVOLUTIONAL-RECURRENT PV POWER FORECASTING, CODED ON EIGHT DIMENSIONS OF EVALUATION PROTOCOL (SECTION II); COUNTS AND PERCENTAGES ARE PER CODED VALUE WITHIN EACH DIMENSION. FULL PER-PAPER CODING WITH VERBATIM SUPPORTING EVIDENCE IS AVAILABLE AT AN ANONYMISED REPOSITORY, LINK WITHHELD FOR REVIEW.

Dimension	Value	n	%
Night hours excluded	not stated	17	63.0
	yes	8	29.6
	no	1	3.7
	partial	1	3.7
Baseline used	own components only	17	63.0
	other ML models	6	22.2
	none	3	11.1
	convex combination	1	3.7
Skill score reported	no	26	96.3
	yes	1	3.7
Weather source	measured	20	74.1
	not stated	4	14.8
	NWP forecast	1	3.7
	none	1	3.7
	reanalysis	1	3.7
Split type	not stated	22	81.5
	chronological	3	11.1
	k-fold	1	3.7
	rolling-origin	1	3.7
Run-to-run variance reported	no	27	100.0
Code available	not stated	26	96.3
	yes	1	3.7
Documented leakage	none	24	88.9
	documented	3	11.1

Rigorous verification practice exists in solar forecasting. Yang et al. [1], writing with more than thirty co-authors, recommend universally reporting the skill score against the optimal convex combination of climatology and persistence. One paper in our sample of 27 meets that standard: Mayer [3] evaluates day-ahead forecasts for fourteen plants against that convex reference, states its daylight filter as a protocol, and holds out a full year used in no part of model selection; its headline hybrid improvement is 5.2 percent in mean absolute error and 1.0 percent in root-mean-square error over an optimised physical model chain, an order of magnitude smaller than improvements typically reported elsewhere in the sample, and measured against a considerably stronger baseline. The standard is achievable and is published in the same journals the surveyed papers appear in; we adopt its reference forecast here and hold the model fixed while varying it.

The difficulty of comparing results across studies with differing evaluation protocols is not specific to solar forecasting; it has been documented across at least 294 papers in 17 scientific fields, together with a taxonomy of the mechanisms involved [4].

III. DATA AND PROTOCOL

A. Data

Experiments use the Desert Knowledge Australia Solar Centre (DKASC) dataset at Alice Springs (23.767°S, 133.867°E, 558 m), an open dataset of ground-mounted PV installations with per-array power measurement and a co-located weather station [5]: every number in this paper is independently checkable against it. Three fixed-mount arrays are used, chosen for differing module technology at comparable scale: array 11 (BP Solar, polycrystalline silicon, 5.0 kW), array 12 (BP Solar, monocrystalline silicon, 5.1 kW), and array 17 (Sanyo, heterojunction, 6.3 kW), all mounted at 20° tilt. The three arrays share one weather station and are not independent replications of one another.

Splits are chronological whole calendar years: training 2011-2013, validation 2014, test 2015, the last touched once, at the end. Training begins in 2011 because array 17 was commissioned in March 2010, and all three arrays share this training window so that cross-array comparisons are not confounded by differing training length.

Evaluation is restricted to hours with solar elevation above 10°, stricter than the 85° zenith filter Yang et al. [1] describe as typical practice, applied for the reason they give: at low solar elevation the clear-sky index becomes unstable relative to a small denominator. One further exclusion is applied: array 17 was switched off for a documented maintenance outage from 5 to 9 June 2015, excluded from evaluation for that array only.

B. The Excluded Array

A fourth array, array 07 (First Solar, cadmium telluride, 7.0 kW), was initially included and subsequently excluded, and is reported rather than omitted, because the mechanism by which it was nearly retained is itself evidence for this harness. A conventional completeness audit passed array 07 for 2014 with 99.99 percent coverage and 0.00 percent missing values in the power channel; the array had in fact produced no power from March to September that year, with 48.41 percent of its daylight hours recording exactly zero output. The audit did not detect this because the logger continued to record faithfully, and what it recorded was zero: zero is not missing, and a completeness-only check is structurally blind to a healthy sensor reporting a dead array. A second audit, testing for near-zero rather than exactly-missing output, is what caught it, and both audits are included in the anonymised repository accompanying this paper.

C. Protocol

We state here only the rules bearing on this paper's claim; the full evaluation protocol, including feature construction and significance testing, is available in the anonymised repository accompanying this paper.

Splits are never shuffled, and all scalars and feature statistics used by the models are fitted on training data only.

Three reference forecasts are used. Smart persistence forecasts $\hat{P}(t) = k_p(t - h) \times P_{cs}(t)$, persisting the clear-sky index from the issue time. Climatology forecasts $\hat{P}(t) =$

$\bar{k}_p(\text{month}, \text{hour}) \times P_{cs}(t)$, with the mean clear-sky index per calendar month and hour of day estimated on training data only. The convex reference is $w \times \text{persistence} + (1 - w) \times \text{climatology}$, with w selected by grid search to minimise root-mean-square error on the validation split, fitted independently for each array, horizon, seed and regime. Following Yang et al. [1], the convex reference is the headline metric; skill against plain persistence is reported alongside because the gap between the two is a result, not a robustness check.

The skill score is $s = 1 - \text{RMSE}_{\text{forecast}} / \text{RMSE}_{\text{reference}}$, computed against both references over the same forecasts and the same evaluation sample.

IV. RESULTS

Figures are for the validation year (2014) unless stated otherwise; held-out test-year (2015) confirmation, evaluated once after every modelling and evaluation choice was frozen, is reported alongside each finding. Table II gives the array-11 configurations underlying both effects in this section; arrays 12 and 17 are reported in the text below.

A. The Reference Forecast Changes Both Magnitude and Shape

Holding the model, the data and the training procedure fixed, we compute skill for the same forecasts against two references: smart persistence and the convex reference. For gradient boosting on array 11 over daylight hours, skill against persistence is 0.252, 0.527 and 0.652 at one, three and six hours; against the convex reference, over the same rows and the same forecasts, it is 0.200, 0.276 and 0.194 (Table II). At six hours, 70 percent of the apparent skill is attributable to the reference. The pattern holds on all three arrays: 0.244 to 0.190, 0.510 to 0.275 and 0.639 to 0.201 on array 12; 0.166 to 0.122, 0.341 to 0.183 and 0.386 to 0.132 on array 17.

The shape, not only the magnitude, changes. Measured against persistence, skill rises monotonically with horizon - the pattern commonly reported as evidence that a method's advantage grows at longer lead times. Measured against the convex reference, skill is non-monotonic and peaks at three hours, falling by six hours to approximately its one-hour value. Fig. 1 shows this for all three arrays: the solid curve (persistence) rises monotonically in every panel; the dashed curve (convex reference) peaks at three hours in every panel and falls back at six.

The mechanism is structural. Smart persistence forecasts $\hat{P}(t) = k_p(t - h) \times P_{cs}(t)$, and at a six-hour horizon a midday target is issued near dawn, when no valid clear-sky index yet exists: the proportion of daylight forecasts relying on a forward-filled rather than directly observed index rises from 3.3 percent at one hour to 22.6 percent at three and 51 percent at six, and the resulting bias reaches -2.15 kW at midday on a 5 kW array. The fitted convex weight w reflects this directly, falling from 0.77 at one hour to 0.25 at three and 0.04 at six on array 11 - at long horizons the optimal reference is almost entirely climatology, which has no comparable staleness problem. Array 17 is a partial exception:

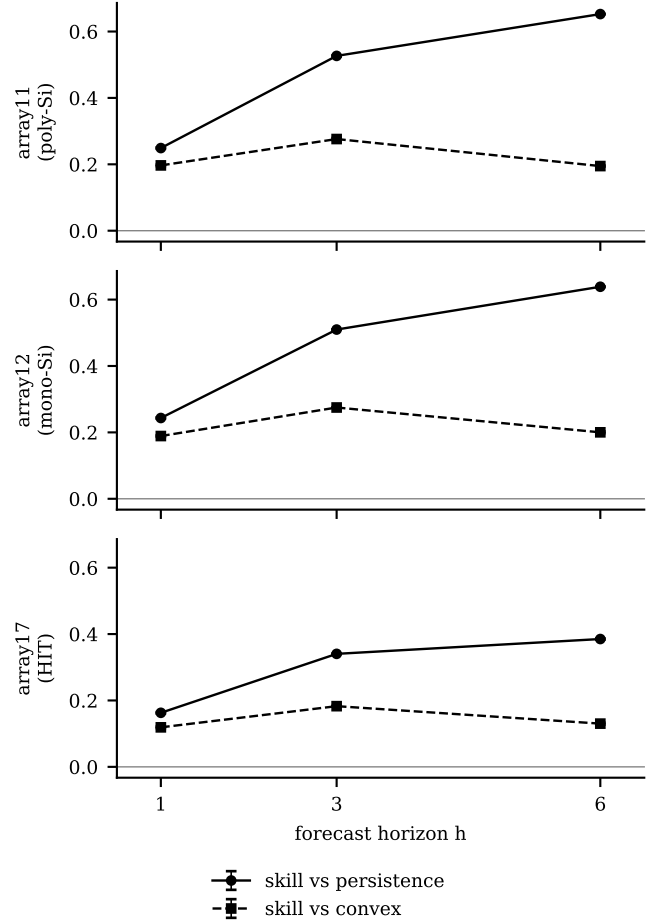


Fig. 1. Forecast skill against two reference forecasts, XGBoost, lagged regime, validation year 2014, one panel per array. Skill against smart persistence (solid) rises monotonically with horizon in every panel; skill against the optimal convex combination of climatology and persistence (dashed) peaks at three hours and falls back at six. The apparent horizon trend is a property of the reference, not the model. Error bars show one standard deviation across five seeds and are smaller than the markers. Only XGBoost is shown; the other four architectures follow the same pattern.

its fitted weight remains higher at every horizon (0.83, 0.50, 0.31), because its persistence forecast is stronger in relative terms, and its non-monotonic peak in Fig. 1 is correspondingly shallower.

Under a paired Diebold-Mariano test with HAC (Bartlett-kernel) variance, the Harvey-Leybourne-Newbold small-sample correction, and Holm-adjusted p-values [6], [7], the convex reference beats smart persistence in all nine array-horizon cells on both the validation and the test split ($p_{\text{holm}} < 0.05$ in all 18 comparisons): the reference this paper treats as proper is not merely a construction, it is a significantly harder target.

The shape inversion itself, not only the significance of the reference gap, replicates on the held-out test year (2015), evaluated once after every modelling and evaluation choice was frozen, on all three arrays. Skill against persistence is 0.266, 0.574 and 0.701 on array 11, 0.263, 0.556 and 0.684 on

TABLE II

PROTOCOL CONFIGURATIONS, ARRAY 11, LAGGED REGIME, VALIDATION YEAR 2014: SAMPLE COUNT, NORMALISED RMSE, SKILL SCORE, AND REFERENCE FORECAST FOR SIX EVALUATION-PROTOCOL CONFIGURATIONS (C1-C6) AT EACH OF THE THREE HORIZONS. C1: DAYLIGHT HOURS ONLY, SKILL AGAINST THE CONVEX REFERENCE (THE PROTOCOL USED THROUGHOUT THIS PAPER). C2: DAYLIGHT HOURS ONLY, SKILL AGAINST SMART PERSISTENCE. C3: HOURS WHERE THE CONVEX REFERENCE HAS A PREDICTION (NOT A FULL 24-HOUR CYCLE - INCLUDED ONLY TO DOCUMENT THIS COVERAGE RESTRICTION). C4: ALL 24 HOURS, SKILL AGAINST SMART PERSISTENCE. C5 AND C6: RAW nRMSE AT DAYLIGHT-ONLY AND ALL-24-HOURS, NO REFERENCE FORECAST. COMPARING C5 AGAINST C6 SHOWS THE CLOSED-FORM NIGHT-HOUR DEFLATION; COMPARING C1 AGAINST C2 SHOWS THE REFERENCE-FORECAST EFFECT - AT $h = 6$, SKILL FALLS FROM +0.652 AGAINST PERSISTENCE TO +0.194 AGAINST THE CONVEX REFERENCE, USING THE IDENTICAL FORECASTS. RESTRICTED TO ARRAY 11; ARRAYS 12 AND 17 SHOW THE SAME PATTERN (SECTION IV).

h	Config	n	nRMSE (%)	skill	Reference
1	C1	3773	6.36	+0.200	convex
	C2	3773	6.36	+0.252	persistence
	C3	4255	6.00	+0.207	convex
	C4	8694	4.20	+0.261	persistence
	C5	3773	6.36	—	—
	C6	8694	4.20	—	—
3	C1	3762	8.97	+0.276	convex
	C2	3762	8.97	+0.526	persistence
	C3	4247	8.44	+0.276	convex
	C4	8694	5.90	+0.528	persistence
	C5	3762	8.97	—	—
	C6	8694	5.90	—	—
6	C1	3756	10.71	+0.194	convex
	C2	3756	10.71	+0.652	persistence
	C3	4245	10.07	+0.194	convex
	C4	8694	7.04	+0.652	persistence
	C5	3756	10.71	—	—
	C6	8694	7.04	—	—

array 12, and 0.077, 0.281 and 0.381 on array 17, at one, three and six hours - monotonic throughout. Against the convex reference it is 0.210, 0.316 and 0.232 on array 11, 0.202, 0.305 and 0.225 on array 12, and 0.043, 0.156 and 0.144 on array 17: non-monotonic on every array, peaking at three hours and falling back by six, on data no evaluation choice was tuned against. Array 17's absolute skill shifts between years by more than the other two arrays' does, and the shift traces to its reference rather than its models: array 17's convex weight leans more heavily on persistence at every horizon (0.31 to 0.83, against 0.04 to 0.77 for arrays 11 and 12), so its reference is structurally more exposed to persistence's year-to-year variation. This affects magnitude, not the shape the paper claims; the full decomposition is in the anonymised repository accompanying this paper.

B. Night Hours Deflate Error, Not Skill

Restricting evaluation to daylight hours or retaining all 24 hours changes reported error substantially and reported skill almost not at all. If night-time errors are approximately zero, then

$$\text{RMSE}_{\text{all}} = \text{RMSE}_{\text{daylight}} \times \sqrt{N_{\text{daylight}}/N_{\text{all}}}. \quad (1)$$

At six hours on array 11, $N_{\text{daylight}} = 3756$ and $N_{\text{all}} = 8694$, giving a predicted ratio of 0.657 against an observed 0.658; agreement is within 0.3 percent at every array and horizon. Reported normalised error at this site is therefore deflated by approximately 34 percent by a protocol choice unreported in 17 of the 27 papers we surveyed (Table I), and the

deflation factor is a property of latitude and season, not of the forecasting method. Skill, by contrast, is nearly invariant to the same choice - for gradient boosting on array 11 it moves from 0.252 to 0.261, 0.527 to 0.528 and 0.652 to 0.652 at one, three and six hours - because the reference forecast is deflated by the same factor as the model.

C. Architecture Contributes Comparatively Little

Across five architectures - gradient boosting, a recurrent network, a convolutional-recurrent network, and both recurrent variants with a gradient-boosted residual correction stage - evaluated under the identical protocol above, the largest difference we measure between any two architectures at any array and horizon is 0.024 in skill score, an order of magnitude smaller than either protocol effect reported above.

D. Held-Out Test Year

The reference-forecast shape inversion, shown above to hold on the test split, is not the only direction that does. Night hours remain deflationary and skill remains nearly unchanged by the same closed-form argument, which does not depend on which year it is applied to, and the largest architecture difference remains an order of magnitude smaller than either protocol effect on the test split as well as on validation.

V. LIMITATIONS

All results come from three co-located arrays at a single desert site, sharing one weather station, so their forecast errors are correlated and the nine array-horizon cells are

not nine independent trials. One evaluation year is used, at horizons of one to six hours. Forecasts are deterministic; no probabilistic evaluation is performed. These results establish that the reference forecast and the treatment of night hours materially change reported accuracy at this site - not that the same magnitudes hold at a different site or climate.

VI. CONCLUSION

Holding the model, the data and the training procedure fixed, we varied only the reference forecast and the treatment of night hours. The reference forecast changes both the magnitude and the shape of reported skill: it accounts for 70 percent of apparent skill at a six-hour horizon and inverts the trend from monotonically increasing to a peak at three hours, a shape inversion that holds on a held-out test year and is backed by significant Diebold-Mariano tests in every array-horizon cell on both splits. Including night hours deflates normalised error by approximately 34 percent, closed-form in the daylight fraction, while leaving skill nearly unchanged. Both effects exceed the largest difference we measure across five architectures under an identical protocol, 0.024 in skill score.

A. Recommendations

Three practices, none of them novel, would make results in this literature comparable. Report a skill score against a stated reference forecast: against smart persistence alone our results appear to improve steadily with horizon, against the convex combination of climatology and persistence recommended by Yang et al. [1] they do not, and the two references support opposite conclusions from identical forecasts. Report the evaluation sample: whether night hours are included, what daylight threshold is applied, and how many samples each cell contains are each sufficient to move a reported error by more than the improvement typically claimed. Report run-to-run variance: none of the 27 papers we surveyed does, and a single run cannot separate a genuine effect from retraining noise.

The harness, the 900 run records, and the coded survey with its supporting evidence are available at an anonymised repository, link withheld for review.

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